Mukesh Patel School of Technology Management & Engineering
Basic Science and Humanities Department

Course Policy Document

Program/Branch/Semester	**:**	B. Tech. (IT, Computer Engineering, Computer Science, EXTC, Mechatronics, Data Science, Cyber Security) /MBA Tech. (IT, Computer, Data Science)
Academic Year	**:**	2025-26
Course Code & Name	**:**	BTET_____Quantum Physics
Credit Details	**:**	L T P C 2 0 2 3
Name of Course Anchor Faculty:	**:**	Dr. Vineet Panchal Vineet.Panchal@nmims.edu
Faculty associated with the course	**:**	Dr. Ajay Phirke Dr. Kuntal Chakrabarti Dr. Tista Basak Dr. Heena Shevde Dr. Arti Hadap Dr. Vinita Khatri Dr. Sreemoyee Sarkar Dr. Vivekanand Bagal Dr. Vishab Kesarwani Dr. Dharmendra Kumar Dr. Ranvir Panwar Dr. Soumyajit Seth Dr. Munindra Jain Dr. Padmshree Patil
Pre-Course Activity: Using GenAI, explore the following: 1. Why to study Quantum Physics? 2. Applications of Quantum computing.		
Course link	**:**	Portal Link MS Teams Link

1 Introduction to the Course

1.1 Importance of the course

Quantum mechanics is the pillar of modern physics. Several current technical advancements like quantum computation, global positioning system, bioinformatics, drug and IC chip design etc. stem directly from quantum physics. Therefore, this topic is important element in the training of an engineering student majoring in diverse disciplines ranging from computer science and related disciplines to biomedical engineering.

1.2 Objective of the Course

This course is aimed to teach the drawbacks of classical physics in explaining several experimental observations and old quantum theory; and to discuss the necessity of new mechanics and the laws related to it. This course will also introduce the concepts of Quantum computing.

1.3 Pre-requisite

Basic knowledge of Physics and Mathematics commensurate with semester I of B Tech / MBA Tech.

2 Course Outcomes (CO)

After completion of the course, students would be able to:

1. Understand the need for Quantum mechanics (K2)
2. Explain the basic laws related to quantum mechanics and apply them to solve related problems. (K3)
3. Apply the concept of quantum mechanics to real-life instrumentation and applications. (K3)
4. Understand the requirement of quantum computation and its various aspects. (K2)
5. Gauge the theoretical principles through laboratory experimentation, enhancing problem solving abilities and technical proficiency in modern physics applications. (K3)

3 Teaching-learning methodologies

3.1 Instruction Plan

Lecture No.	Topic	Teaching Method	Blooms Level	Resources	COs Mapped	Assessment and Evaluation
Unit 1:						
1	Introduction to course and Course Policy. Introduction to Quantum Physics.	Lecture	Understand	TB1	CO 1	Test-1, TEE

2	Black body radiation, Explanation of it using the photon concept	Lecture, Practical	Understand, Apply	TB1	CO 1	Test-1, Lab submission, Practical Exam, TEE
3	Photoelectric effect, Compton effect, de Broglie hypothesis		Understand, Apply	TB1	CO 1	
4	Experiments demonstrating wave properties of electron: Electron interference (double slit experiment), Electron Diffraction (Davison - Germer experiment), Uncertainty Principle.		Understand, Apply	TB1	CO 1, CO2	
5	Wave-particle duality, Born's interpretation of the wave function, Verification of matter waves.		Understand, Apply	TB1	CO 1, CO2	
6	Numerical on Unit 1.		Apply	TB1	CO1, CO2, CO5	Test 1, TEE
Unit 2:						
7	Basic postulates of quantum mechanics, concept of wave function, Superposition principle of eigenstates.	Lecture,	Understand	TB1	CO2	Test-1, TEE
8	Born's interpretation of the wave function, Superposition principle of eigenstates.	Lecture	Understand	TB1	CO2	Test 1, TEE
9	Concept of collapse of wave function. Time dependent and time independent Schrodinger Equation,	Lecture	Understand, Apply	TB1	CO2	Test 1, TEE
10	Class Test 1					
11	Concept of free particle	Lecture	Understand, Apply	TB1	CO2	Test 2, TEE
12	Particle in an infinite potential well.	Lecture	Understand, Apply	TB1	CO2	Test-2, TEE
13	Particle in a finite potential well.	Lecture	Understand, Apply	TB1	CO2	Test-2, TEE

14	Box problem. Bound vs. unbound states.			TB1	CO2	Test-2, TEE
15	Numerical problems on Unit 2.	Lecture	Apply		CO2, CO5	Test 2, TEE
Unit 3						
16	Concept of Quantum Tunnelling.	Lecture, Practical	Understand, Apply	TB1	CO2	Lab submission, Practical Exam, TEE
17	Reflection and Transmission coefficients	Lecture	Understand	TB1	CO2	TEE, Test 2
18	Reflection and Transmission coefficients	Lecture	Understand	TB1	CO2	TEE, Test 2
19	Few realistic examples of tunnelling, e.g., alpha decay, Probe microscopes (Scanning Tunnelling microscope).	Lecture, Practical	Understand, Apply	TB1	CO3	Lab submission, Practical Exam, Test 2, TEE
20	Simple Harmonic Oscillator, explanation in 1D (no detailed derivation).		Understand, Apply	TB1	CO3	
21	Hydrogen atom	Lecture	Understand, Apply	TB1	CO3	TEE
22	Numerical problems on Unit 3.	Lecture	Apply		CO3, CO5	Test 2, TEE
23	Class Test 2					
Unit 4						
24	Limitations of classical computation	Lecture,	Understand	TB3	CO4	TEE
25	Difference between classical and quantum computer,	Lecture,	Understand, Apply	TB3	CO4	TEE
25	Qubits.	Lecture, Practical	Understand, Apply	TB3	CO4	Lab submission, TEE
26	Bloch Sphere	Lecture	Understand	TB3	CO4	TEE
27	Single Qubit Operations	Lecture, Practical	Apply	TB3	CO4	Lab submission, TEE
28	Quantum gates	Lecture, Practical	Understand	TB3	CO4	Lab submission, TEE
29	Bell States	Lecture	Understand	TB3	CO4	TEE
30	Quantum Entanglement.	Lecture	Understand	TB3	CO4	Lab submission, TEE

3.2 Pre-class activity

Preliminary study material in different forms will be made available on the student portal / MS Teams of the class. Students are expected to go through this material before attending the upcoming session. It is expected that the students put in at least two hours of self-study for every one hour of classroom teaching. During the lecture session, more emphasis will be given on in-depth topics, practical applications and doubt solving.

3.3 References

Textbook

1. A. Beiser, S. Mahajan and S. Choudhury, "Concept of Modern Physics", Tata McGraw Hill, 7th edition (SIE) 2015.
2. Arthur Beiser, ''Perspectives of Modern Physics'', McGraw Hill, 1969
3. Bernhardt, Chris, Quantum computing for everyone, MIT Press, 2020.
4. McMahon, David, Quantum computing explained, John Wiley & Sons, 2007.

Reference Books

1. Eisberg and Resnik, "Quantum Physics of Atoms, Molecules, Solids, Nuclei and Particles" Wiley, 2nd edition 2006.
2. R. A. Serwey, C. J. Moses, C. A. Moyer, "Modern Physics", Thomson, 3rd edition 2005.
3. David J. Griffiths, "Introduction to Quantum Mechanics", Pearson, 2nd edition 2015.
4. Robert S. Sutor, "Dancing with Qubits How quantum computing works and how it can change the world", Packt Publishing, 2019

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4 Assessment and Evaluation Scheme

Assessment Component	Lab Performance	Lab Test	Poster presentation	Assignments	Class performance	Class Test-1 and Class Test-2	TEE (100 marks) (Marks scaled to 50)
Marks	10	5	5	5	5	10+10	100
Date/week of activity	Weekly	Week 9	Week 7	Respective faculty will decide the week.	Week 3 and week 8	Week 5 and Week 10	

4.1 Internal Continuous Assessment (ICA) – 50 marks

Assessment of ICA comprises the following components.

1. Class test 1 and 2

a. Two class tests will be conducted as per the academic calendar.
b. It will be conducted offline for 10 marks each.

2. Lab performance and class performance evaluation (10 + 5 marks)

i. Continuous assessment for laboratory experiments will be conducted. There are 10 experiments in the course.
ii. Discussion of your work with your peers is allowed. However, each student is expected to submit his/her original work. Assessment of the lab work will be carried out based on parameters such as timely completion of lab work by the student, understanding of the experiment performed, regularity, discipline etc. during the session.
iii. Class performance assessment is to ensure the active involvement of students during the lectures. The idea is to encourage students to pay attention in class and actively participate.

3. Lab Test (5 marks)

Practical exam will be conducted based on the experiments done.Viva will be conducted as a part of the lab exam.

4. Poster (5 marks)

a. Faculty will make group of 4 students.
b. Faculty will explain the idea behind choosing posters.
c. Get the topic approval.
d. Assessment will be based on the report, presentation, and discussion.

5. Assignments

In-class or home assignments will be given, and carry a weightage of 5 marks. Evaluation is based on timely submission. 50% marks would be deducted for late submissions.

4.2 Term End Examination (TEE) – (100 marks scaled down to 50)

TEE conducted at the end of the semester will be for 100 marks (3-hour duration). Marks obtained scaled down to 50. There will be 7 questions (each question will be 20 marks). Q1 will be compulsory and any 4 from the remaining to be solved.

4.3 Course Passing Criteria

4.3.1 ICA (50 marks) – No minimum marks

4.3.2 TEE (100 marks scaled to 50) – 40% required for passing

4.3.3 (ICA + TEE) (100 marks) - 40% required for passing

5 Laboratory details

The 10 experiments from the following list of the experiments and one poster will form the submission for laboratory coursework.

Sr. No.	Week No.	List of Lab Experiments	Mapped CO
1.	Week 1 & Week 2	To study the photoelectric effect with change in frequency and to find stopping potential.	CO5
2.		To perform the Franck-Hertz experiment.	CO5
3.	Week 3 & Week 4	Determination of Stefan's constant σ.	CO5
4.		To study the photoelectric effect with intensity.	CO5
5.	Week 5 & Week 6	Determination of Boltzmann's constant.	CO5
6.		Determination of Planck's constant.	CO5
7.	Week 7 & Week 8	Presentation of poster	CO5
8.		To study the characteristics of tunnel diode.	CO5
9.	Week 9 & Week 10	Study of universal quantum gates and their measured outcomes.	CO5
10.		Understanding concept of quantum superposition using the example of quantum coin.	CO5
11.	Week 11 & Week 12	Design of circuits using quantum gates to demonstrate quantum entanglement phenomena.	CO5
12.		Practice session of practical.	CO5
13.	Week 13	Practical Exam	CO5
14.	Week 14	Practical Exam	CO5
15.	Week 15	Viva	CO5

6. **Tutorial Plan** This course does not have any tutorial.

7. **Course Material**: References and Lab Manuals would be uploaded on LMS by faculty every week.

8. **GenAI Usage**: GenAI will be used in tailoring instruction to individual learner needs, providing personalized feedback, and creating interactive learning environments.

9. **Academic Integrity Statement**: Students are expected to carry out assigned work under Internal Continuous Assessment (ICA) independently. Copying in any form is not acceptable and will invite strict disciplinary action. Evaluation of corresponding components will be affected proportionately in such cases. Plagiarism detection software will be used to check plagiarism wherever applicable. Academic integrity is expected from the students in all components of course assessment.